Complexified Cube Chains and a Braid Invariant of Precubical Sets

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Abstract

Ziemiański's cube chain category $\mathrm{Ch}(K)$ is a combinatorial model for the space of directed paths on a precubical set K. Paliga and Ziemiański built a braid invariant for precubical sets by observing a relation between $\pi_1|\mathrm{Ch}(Z)|$ and braids, where Z is the terminal precubical set. However, that invariant trivializes for Euclidean K because $|\mathrm{Ch}(I^n)|$ is contractible [PZ21, Prop. 6.1]. Following an analogy from the hyperplane arrangement literature, we build a "complexified" version of Ch, which we call Ch^*. It does not collapse: in fact $\mathrm{Ch}^*(I^n) \cong \mathrm{Sal}(n)^{\mathrm{op}}$, the Salvetti poset of the braid arrangement. Salvetti's theorem tells us $\pi_1|\mathrm{Ch}^*(I^n)|$ is isomorphic to the pure braid group P_n. The construction of Ch^* works by attaching one dimensional refinements to each cube chain in K. Furthermore, we associate permutations with each morphism, which give us a functor $\mathrm{Conc} : \mathrm{Ch}^*(K) \to \mathbf{Braid}^+$ into the positive braid monoid. It is natural in K, and non-trivial as long as K contains a 2-cell. The construction and the comparison with $\mathrm{Sal}(n)$ are formalized in Lean [Sto26].

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1 Introduction

1.1 Background

Ziemiański's cube chain category $\mathrm{Ch}(K)$ is a combinatorial model for the executions of a bipointed precubical set K [Zie20, Thm. 7.6]. Paliga and Ziemiański established a homotopy equivalence between $|\mathrm{Ch}(Z)|$ and configuration spaces of points in the plane, where Z is the terminal precubical set. It extends, by naturality, to an invariant of any precubical set K. However, it trivializes on any Euclidean complex, see [PZ21, Prop. 6.1]. The proof is one line: the invariant factors through $|\mathrm{Ch}(I^n)|$, which is contractible.

The key observation came through an attempt to formalize morse flows for precubical sets. Consider the space of "execution times" as a vector in $\mathbb{R}^n$, where each coordinate is an event, and the values in the vector represent when that event occurred. The association of vectors with cube chains is precisely the faces of the braid arrangement. The trivialization phenomenon also occurs for hyperplane arrangements. However, that literature has a solution: consider analogous hyperplane arrangements in $\mathbb{C}$ instead of $\mathbb{R}$. This results in the Salvetti poset [Par12], whose elements are pairs of faces and minimal faces refining them. Despite discovering neither a suitably generic morse framework, nor a notion of "complexified precubical sets", the construction of the salvetti complex generalizes well.

The minimal faces of an arrangement are called chambers or topes. In the precubical setting, these correspond to one dimensional cube chains, we call *runs*. So we push this analogy through, and build Ch^* as a set of pairs, whose first component is an element of Ch, and second component is a run refining it. The construction $\mathrm{Ch}^*(K)$ works for any precubical set, and is indeed natural in K. Furthermore we find the analogy with hyperplane arrangements is precise for cubes:

Theorem 1.1 (Theorem 5.11)**.** *For every $n \geq 0$ there is an isomorphism of posets $\mathrm{Ch}^*(I^n) \cong \mathrm{Sal}(n)^{\mathrm{op}}$, where $\mathrm{Sal}(n)$ is the Salvetti poset of the braid arrangement A_{n-1}. Consequently $|\mathrm{Ch}^*(I^n)| \simeq \mathrm{Conf}(n, \mathbb{C})$ and $\pi_1|\mathrm{Ch}^*(I^n)| \cong P_n$, the pure braid group on n strands.*

So indeed this circumvents the issue with Euclidean complexes from [PZ21, Prop. 6.1]. Furthermore, the braid data can also be extracted directly from morphisms of $\mathrm{Ch}^*(K)$. No need to pass through Salvetti's theorem or nerves at all. The morphisms contain permutation data that allows us to build braid words in the Garside germ presentation [Deh+15]:

Theorem 1.2 (Theorem 6.6, Proposition 6.7)**.** *For every bipointed precubical set K there is a functor $\mathrm{Conc} : \mathrm{Ch}^*(K) \to \mathbf{Braid}^+$ into the positive braid monoids, sending a morphism to the germ of the permutation by which it reorders events. It is natural in K, and it is nontrivial if K contains a 2-cell.*

1.2 Formalisation

Everything above except the topology is formalised in Lean 4 on top of mathlib [Sto26]. It also builds the foundational precubical theory, the necessary results from [PZ21]. Some formalisations are slightly modified from the text, but those deviations are noted. It also provides a verified computation for Conc.

1.3 Notation

To fix notation, let **Cube** be the box category: its objects are the cubes I^n for $n \geq 0$, and its morphisms are the face embeddings (composites of the coface maps δ_i^ϵ). A *precubical set* is a presheaf on **Cube**; write $\square$ for the category of precubical sets. Concretely, a precubical set K has, for each $n \geq 0$, a set K_n of n-cells, together with face maps

$$d_i^\epsilon : K_n \to K_{n-1} \qquad (1 \leq i \leq n,\ \epsilon \in \{0, 1\}),$$

where i names a direction and ϵ a side, subject to the precubical identities $d_i^\epsilon d_j^\eta = d_{j-1}^\eta d_i^\epsilon$ for $i < j$. We write I^n both for the standard n-cube as an object of **Cube** and for the presheaf it represents, and we set $[n] := \{1, \dots, n\}$, so that $[0] = \emptyset$. Yoneda identifies the two descriptions of a cell,

$$K_n \;=\; K(I^n) \;=\; \mathrm{Hom}_\square(I^n, K),$$

and we pass between them without comment; in particular K_0 is the set of vertices.

We usually work with precubical sets carrying designated start and end vertices. A *bipointed* precubical set is a precubical set K with two chosen vertices $\mathrm{init}_K, \mathrm{fin}_K \in K_0$, and morphisms are required to preserve them; write $\square_*$ for the resulting category. Each cube I^n is bipointed by its initial vertex $(0, \dots, 0)$ and final vertex $(1, \dots, 1)$. We write Z for the final object of $\square_*$.

Convention 1.3. Named ambient categories are set in bold (**Cat**, **Set**, **Cube**, **Braid**$^+$); constructions applied to an object are set upright ($\mathrm{Ch}(K)$, $\mathrm{Run}(K)$, $\mathrm{Sal}(n)$). Functors are written with $\to$; we reserve $\Rightarrow$ for natural transformations. The standard n-cube is I^n; the symbol $\square$ names the category of precubical sets.

Remark 1.4 (Variance). The $^{\mathrm{op}}$ in Theorem 1.1 is merely an artefact of choices made by different literature we reference. That is, in the cube chain literature [PZ21; Zie20] a morphism of $\mathrm{Ch}(K)$ runs from a *finer* chain to a *coarser*. And in the arrangement literature [Par12] the face order runs from *smaller* faces to *larger* ones.

2 Background: cube chains

We recall two constructions on $\square_*$ from [PZ21; Zie20].

The first is the **wedge** $\vee$. Given $P, Q \in \square_*$, the wedge $P \vee Q$ is the pushout, formed in $\square$, gluing P's final vertex to Q's initial vertex:

$$\begin{tikzcd} I^0 \arrow[r, "\mathrm{init}_Q"] \arrow[d, "\mathrm{fin}_P"'] & Q \arrow[d] \\ P \arrow[r] & P \vee Q \arrow[ul, phantom, "\ulcorner", very near start] \end{tikzcd}$$

Note that the two legs are maps of $\square$, not of $\square_*$. The wedge $P \vee Q$ is bipointed by init_P and fin_Q, and $(\square_*, \vee, I^0)$ is monoidal with unit the point I^0. The **serial wedge** of a list of *positive* naturals is

$$\bigvee[a_1, \dots, a_r] := I^{a_1} \vee \cdots \vee I^{a_r},$$

with $\bigvee[\,] = I^0$. Write $\square^\vee \subseteq \square_*$ for the full subcategory of serial wedges.

The second construction is $\mathrm{Ch}(K)$. For $K \in \square_*$, let $\mathrm{Ch}(K)$ be the full subcategory of the slice $\square_*/K$ spanned by the serial wedges: an object is a bipointed map $\bigvee a \to K$, and a morphism

$(\bigvee a \to K) \to (\bigvee b \to K)$ is a bipointed map $\bigvee a \to \bigvee b$ commuting over K. Such a morphism runs from a finer chain to a coarser one.

Finally, two properties of K from [PZ21] recur. We say K *admits altitude* if there is a function $\mathrm{alt} : K_0 \to \mathbb{Z}$ with $\mathrm{alt}(\mathrm{fin}(e)) = \mathrm{alt}(\mathrm{init}(e)) + 1$ for every edge $e \in K_1$. Following [PZ21, §2] verbatim, K is *non-self-linked* (NSL) if every $\square$-map $I^n \to K$ is injective, for all $n \geq 0$.

We use the following, all from [PZ21].

Lemma 2.1 ([PZ21, §2])**.** *The $\vee$ operator is a monoidal product on $\square_*$, with unit I^0.*

Lemma 2.2. $\mathrm{Ch} : \square_* \to \mathbf{Cat}$ *is a lax monoidal functor from* $(\square_*, \vee)$ *to* $(\mathbf{Cat}, \times)$*: the comparison* $\mathrm{Ch}(P) \times \mathrm{Ch}(Q) \to \mathrm{Ch}(P \vee Q)$ *concatenates chains.*

Lemma 2.3. *When K admits altitudes and is NSL, the functor of Lemma 2.2 is an isomorphism of categories, since both categories are skeletal and posets. This is an easy corollary of [PZ21, Lem. 2.11]*

Lemma 2.4. *Serial wedges admit altitude and are NSL. And specifically so the final vertex of $\bigvee a$ has altitude $\sum_i a_i$.*

Lemma 2.5 ([PZ21, Prop. 2.7])**.** *Let P, Q be connected bipointed precubical sets admitting altitude functions. Then every bipointed map $f : P \to Q$ preserves altitude:* $\mathrm{alt}_Q \circ f = \mathrm{alt}_P$*.*

This is all the foundational material we need.

3 Wedge machinery

Most of the technical material in this paper involves bookkeeping around coordinates of wedges. So we build a bit of machinery here to assist.

Lemma 3.1. *A functor* $P : \mathbf{Cube} \to \mathbf{Set}$ *extends, by left Kan extension along the Yoneda embedding, to a cocontinuous* $E(P) : \square \to \mathbf{Set}$ *with* $E(P)(I^n) = P(I^n)$*. If* $P(I^0) = \emptyset$ *then for every serial wedge*

$$E(P)\Big(\bigvee a\Big) \;\cong\; \coprod_i P(I^{a_i}). \tag{1}$$

Proof. Applying $E(P)$ to the pushout defining $A \vee B$ gives $E(P)(A \vee B) = E(P)(A) \sqcup_{P(I^0)} E(P)(B)$, which is the coproduct when $P(I^0) = \emptyset$. $\square$

Lemma 3.2. *Let P be a precubical set and put* $E(P) := \mathrm{Hom}_{\square}(-, P) : \square^{\mathrm{op}} \to \mathbf{Set}$*, the right Kan extension of P along Yoneda. If P_0 is a singleton then for every serial wedge*

$$E(P)\Big(\bigvee a\Big) \;\cong\; \prod_i P_{a_i}. \tag{2}$$

Proof. A hom functor takes colimits in its first variable to limits, so $E(P)$ sends the pushout defining $A \vee B$ to the pullback $E(P)(A) \times_{E(P)(I^0)} E(P)(B)$. Here $E(P)(I^0) = P_0$ is a singleton, hence terminal in $\mathbf{Set}$, and a pullback over a terminal object is a product. $\square$

Definition 3.3. Let Coord : **Cube** $\to$ **Set** send the cube I^n to its set of directions $[n]$, and send a face embedding $g : I^n \to I^m$ to the monotone injection $g_* : [n] \hookrightarrow [m]$ that names the n directions g leaves free. Since $\mathrm{Coord}(I^0) = [0] = \emptyset$, Lemma 3.1 applies:

$$E(\mathrm{Coord})\Big(\bigvee a\Big) \;\cong\; \coprod_i [a_i].$$

For a morphism $f : \bigvee a \to \bigvee b$ we write $f_* := E(\mathrm{Coord})(f)$.

Definition 3.4. Write $\mathrm{blk}(c)$ for the summand index of a coordinate $c \in \coprod_i [a_i]$. By the pushout defining $\bigvee a$, a morphism $f : \bigvee a \to \bigvee b$ is determined by the $a_i \to \bigvee b$. But each such image lies in a single I^{b_j} since $a_i > 0$. So it corresponds to a order preserving face embedding

$$f_i : I^{a_i} \longrightarrow I^{b_{\mathrm{blk}(f)(i)}}.$$

This defines $\mathrm{blk}(f)$, and describes f_* on coordinates:

$$f_*(i, d) \;=\; \big(\mathrm{blk}(f)(i),\; (f_i)_*(d)\big), \qquad d \in [a_i]. \tag{3}$$

In particular $\mathrm{blk} \circ f_* = \mathrm{blk}(f) \circ \mathrm{blk}$.

Lemma 3.5. *For any morphism $f : \bigvee a \to I^m$ in $\square$, the function f_* is injective.*

Proof. Induct on a. The empty case is $f_* : [0] \to [m]$, trivially injective. For the induction case, $f : (\bigvee a) \vee I^n \to I^m$, which by the pushout splits into $g : \bigvee a \to I^m$ and $h : I^n \to I^m$ with $f_* = g_* \sqcup h_*$ by (Lemma 3.1). Here g_* is injective by the induction hypothesis, and h_* is the monotone injection for the face embedding h.

So it suffices that $\operatorname{im} g_*$ and $\operatorname{im} h_*$ are disjoint. Let $c \in \{0,1\}^m$ be the image of the glued vertex $\mathrm{fin}(\bigvee a) = \mathrm{init}(I^n)$. Coordinates only increase along a map into a cube, so every direction g leaves free is already 1 at c, while every direction h leaves free is still 0 at c. $\square$

Lemma 3.6. *For any morphism $f : \bigvee a \to \bigvee b$ in $\square_*$, the function f_* is bijective and* $\mathrm{blk}(f)$ *is monotone.*

Proof. To prove injective and monotone, we go by induction on the length of b, generalising over a. If b is empty, then so is a by Lemma 2.5 and Lemma 2.4. So both $\mathrm{blk}(f)$ and f_* are the empty function.

Now consider $f : \bigvee a \to I^m \vee \bigvee b$. By Lemma 2.4 and Lemma 2.3, $\mathrm{Ch}(I^m) \times \mathrm{Ch}(\bigvee b) = \mathrm{Ch}(I^m \vee \bigvee b)$. We can view f as an object of $\mathrm{Ch}(I^m \vee \bigvee b)$. So a splits as $a = a_l * a_r$, and f splits as

$$(f_1 : \bigvee a_l \to I^m, f_2 : \bigvee a_r \to \bigvee b)$$

Since $f = f_1 \vee f_2$, by (Lemma 3.1) we have

$$f_* = (f_1 \vee f_2)_* = (f_1)_* \sqcup (f_2)_*$$

Now $(f_2)_*$ is injective by the induction hypothesis and $(f_1)_*$ is injective by Lemma 3.5. This is a disjoint union of injections with disjoint codomains, hence f_* is injective. For blk, note

$$\mathrm{blk}(f) = \mathrm{blk}(f_1) \sqcup (1 + \mathrm{blk}(f_2))$$

Now $\mathrm{blk}(f_1)$ is always 1, and $1+\mathrm{blk}(f_2)$ is monotone by induction, and > 1. So $\mathrm{blk}(f)$ is monotone as well.

For surjectivity of f_*, we count. By Lemma 2.4 the final vertex of $\bigvee a$ has altitude $\sum_i a_i$, and likewise for b; by Lemma 2.5 the bipointed map f preserves altitude so $\sum_i a_i = \sum_j b_j$. So f_* is an injective function between finite sets with equal cardinality. So f_* is a bijection. □

4 The braid arrangement and the Salvetti poset

The construction of Ch^* ports the following technique to precubical sets.[1] We describe the technique here, following Paris [Par12], and build Ch^* in the next section.

Let $\mathrm{Face}(n)$ be the set of total preorders on $[n]$. A total preorder F, induces an equivalence relation $\sim_F$, which are linearly ordered by $<_F$. The classes of $\sim_F$ are called blocks, which give an ordered set partition of $[n]$. Order $\mathrm{Face}(n)$ by

$$F \leq G \quad := \quad <_F \ \subseteq \ <_G,$$

This is the poset of **covectors**, or faces, of the braid arrangement $A_{n-1} = \{\, x_i = x_j \,\}_{i<j}$ in $\mathbb{R}^n$. A point x gives the covector $i \leq_F j := x_i \leq x_j$, the weak order of its coordinates. The maximal covectors are those with all distinct points, that is the linear orders on $[n]$; these are the **topes**, and they correspond to the components of $\mathbb{R}^n \setminus \bigcup A_{n-1}$.

Definition 4.1. The **Salvetti poset** is

$$\mathrm{Sal}(n) := \{\, (F, C) \ \mid \ F \in \mathrm{Face}(n),\ C \text{ a tope},\ F \leq C \,\}.$$

To define the order on Sal, we first need to define composition. Let F and G be faces, and define $F \circ G$ by

$$i \leq_{F \circ G} j \iff i <_F j \text{ or } \big(i \sim_F j \text{ and } i \leq_G j\big).$$

In particular $F \leq F \circ G$, and $F \circ C$ is again a tope whenever C is. We give $\mathrm{Sal}(n)$ a partial order structure

$$(F, C) \preceq (F', C') \quad :\iff \quad F \leq F' \ \text{ and } \ C' = F' \circ C.$$

This is the order of [Par12, §3.1].

Theorem 4.2 (Salvetti [Par12])**.** $|\mathrm{Sal}(n)| \simeq \mathrm{Conf}(n, \mathbb{C}) = \{z \in \mathbb{C}^n : z_i \neq z_j \text{ for } i \neq j\}$*. This is the ordered configuration space on n points, as well as the complement of the complexified A_{n-1}. So $\pi_1|\mathrm{Sal}(n)| \cong P_n$, the pure braid group.*

The packaging we will imitate is as a Grothendieck construction. Let

$$\mathrm{Tope} : \mathrm{Face}(n) \to \mathbf{Set}, \qquad \mathrm{Tope}(F) := \{C \text{ a tope} : F \leq C\},$$

be the covariant functor sending $F \leq F'$ to the map $C \mapsto F' \circ C$. Then

$$\mathrm{Sal}(n) \ \cong \ \textstyle\int \mathrm{Tope},$$

the Grothendieck construction: its objects are the pairs (F, C) with $C \in \mathrm{Tope}(F)$, and its morphisms are exactly the relations $\preceq$ above. Intuitively, each face is annotated with a compatible tope, a local orientation; crossing a face can flip the orientation, and these flips are the braid letters.

[1] The Lean repo uses Conditional Oriented Matroids instead of ordered set partitions to define the braid arrangement, but proves they are equivalent

5 The complexification Ch^*

The first observation that suggests a bridge is feasible here is that

Lemma 5.1. $\mathrm{Ch}(I^n) \cong \mathrm{Face}(n)^{\mathrm{op}}$

By Lemma 3.6 a chain of I^n is just an ordered set partition of its directions, and a morphism of chains is a refinement. The rest is rote, and formalized in Lean. But, to establish an analogy that works for any precubical set, we need to produce a precubical notion of tope.

5.1 Runs

Definition 5.2. A serial wedge is **one dimensional** if all its entries are 1.

Definition 5.3. Let $\mathrm{Run}(K)$ be the set[2] of one dimensional chains in $\mathrm{Ch}(K)$.

Lemma 5.4. *Restriction to summands is a bijection* $\mathrm{Run}(\bigvee a) \cong \prod_i \mathrm{Run}(I^{a_i})$.

Proof. Induct on the length of a. The empty case is trivial. Now, suppose $\bigvee a = I^{a_1} \vee \bigvee a'$. Given runs $f : \bigvee[1^n] \to I^{a_1}$ and $g : \bigvee[1^m] \to \bigvee a'$, the wedge $f \vee g : \bigvee[1^{n+m}] \to \bigvee a$ is again a run. Conversely, a run $h : \bigvee[1^n] \to \bigvee a$ splits by Lemma 2.3 into chains $h_l : \bigvee c \to I^{a_1}$ and $h_r : \bigvee d \to \bigvee a'$; since $\bigvee c \vee \bigvee d = \bigvee[1^n]$, every entry of c and d is 1, so h_l and h_r are runs. The two constructions are mutually inverse. $\square$

Now, we want a contravariant operation from $\mathrm{Ch}(K)$ to runs, mirroring the order from Salvetti:

$$(\bigvee a \to \bigvee b) \;\longmapsto\; (\mathrm{Run}(\bigvee b) \to \mathrm{Run}(\bigvee a)).$$

We will define a functor $\rho : \mathbf{Cube}^{\mathrm{op}} \to \mathbf{Set}$ which will give us this shape. First, note that every morphism $f : I^n \to I^m$ is a face embedding. So has a corresponding projection on cells $\hat{f} : I^m \to I^n$. Notably, the projection is monotonically decreasing on cell dimension. With that, we can define ρ as follows

Definition 5.5. On objects, let $\rho(I^n) = \mathrm{Run}(I^n)$. By the pushout defining the wedge, a run of I^m is a list $[c_1, \ldots, c_m]$ of composable edges from init to fin. For a face embedding $f : I^n \to I^m$, let

$$\rho(f)[c_1, \ldots, c_m] \;:=\; [\,\hat{f}(c_1), \ldots, \hat{f}(c_m)\,] \quad \text{with the 0-cells deleted.}$$

This is again a run: $\hat{f}(c_k)$ will preserve exactly n coordinates, giving n cells. The projection preserves boundary maps, so even after deleting 0 cells, it remains a wedge map. And $\hat{f}$ carries init to init and fin to fin.[3]

In coordinates, $r_*(k)$ is the free direction of c_k, so the surviving steps are those in $S := r_*^{-1}(\mathrm{im}\, f_*)$, and they keep their relative order. Writing $\iota : [n] \to S$ for the increasing bijection,

$$\rho(f)(r)_* \;=\; f_*^{-1} \circ r_* \circ \iota. \tag{4}$$

Lemma 5.6. $\rho : \mathbf{Cube}^{\mathrm{op}} \to \mathbf{Set}$ *is a functor and* $\rho(I^0)$ *is a singleton; so* ρ *is a precubical set with a single vertex.*

[2] In lean, we make Run a subcategory of $\mathrm{Ch}(K)$, so the bijection becomes monoidality explicitly. This makes no difference to the result.

[3] The lean defines ρ via a slight detour with cube chains instead of wedge maps to avoid some issues with "Fin". But the approaches are proven equivalent

Proof. The projection of the identity embedding is the identity. And projections compose to projections, so ρ is a functor. The empty wedge map is the sole member of $\rho(I^0)$. $\square$

So ρ is a precubical set, and we write $E(\rho) := \mathrm{Hom}_{\square}(-, \rho) : \square_*^{\mathrm{op}} \to \mathbf{Set}$.

Lemma 5.7 (ρ classifies runs)**.** *For every serial wedge $\bigvee a$ there are bijections*

$$E(\rho)(\bigvee a) \;=\; \mathrm{Hom}_{\square}(\bigvee a,\; \rho) \;\cong\; \textstyle\prod_i \mathrm{Run}(I^{a_i}) \;\cong\; \mathrm{Run}(\bigvee a).$$

Proof. The left congruence is Lemma 3.2. The right is Lemma 5.4. $\square$

We will freely identify $E(\rho)(\bigvee a)$ with $\mathrm{Run}(\bigvee a)$.

5.2 The construction

Definition 5.8. Let $W(K) : \mathrm{Ch}(K) \to \square^{\vee}$ be the forgetful functor $(\bigvee a \to K) \mapsto \bigvee a$, natural in K: for $u : K \to L$ we have $W(L) \circ \mathrm{Ch}(u) = W(K)$. Set

$$\mathrm{Lines}(K) := E(\rho) \circ W(K)^{\mathrm{op}},$$

a presheaf on $\mathrm{Ch}(K)$, and

$$\mathrm{Ch}^*(K) := \textstyle\int \mathrm{Lines}(K),$$

the category of elements of $\mathrm{Lines}(K)$, with projection $\pi : \mathrm{Ch}^*(K) \to \mathrm{Ch}(K)$.

We will see that defining things abstractly can some conveinences later. But first let us unpack the definition of Ch^* to have a more concrete presentation of it. Objects of $\mathrm{Ch}^*(K)$ are pairs

$$(\bigvee a \to K,\; x \in \mathrm{Run}(\bigvee a)), \;\text{ i.e. }\; \bigvee[1,\dots,1] \to \bigvee a \to K,$$

a cube chain together with a linear ordering of its events. The intuition here matches the Salvetti story precisely: we have a wedge map into K, coupled with compatible orientation information. Morphisms are a little uglier. A morphism

$$(\bigvee a \to K,\; x \in \mathrm{Run}(\bigvee a)) \longrightarrow (\bigvee b \to K,\; y \in \mathrm{Run}(\bigvee b))$$

unpacks to

1. A morphism $f : (\bigvee a \to K) \to (\bigvee b \to K)$, i.e. a bipointed map $f' : \bigvee a \to \bigvee b$ commuting over K.
2. The condition $x = \mathrm{Lines}(f)(y)$: we take the run y, project it backwards along $\hat{f}$, and require the result to be x.

The diagram below records the story. The solid arrows are morphisms of $\mathrm{Ch}^*(K)$; the dashed lines are not. $E(\rho)(f)$ is the projection associated with f. And $\sigma(f)$ is an induced permutation we will study in the next section

$$\begin{tikzcd}
\bigvee[1,\dots,1] \arrow[r, "x"] \arrow[d, dashed, "\sigma(f)"'] & \bigvee a \arrow[d, "f"] \arrow[dr] & \\
\bigvee[1,\dots,1] \arrow[r, "y"'] & \bigvee b \arrow[r] \arrow[u, dashed, bend left, "E(\rho)(f)"] & K
\end{tikzcd}$$

5.3 The crossing permutation

Let's look more closely at how σ is constructed. Let $N = \Sigma_i a_i = \Sigma_j b_j$. Then define

$$\sigma(f) := y_*^{-1} \circ f'_* \circ x_*$$

and observe that by Lemma 3.6, $\sigma(f)$ is a composition of bijections, so $\sigma(f) \in \Sigma_n$. Also, the form $y_* \circ \sigma(f) = f'_* \circ x_*$ will be convenient. This permutation $\sigma(f)$ is what will power our braid functor.

Lemma 5.9. *$\sigma : \mathrm{Ch}^*(K) \to \bigsqcup_N \Sigma_N$ is a functor, where the target is the groupoid with objects the naturals and $\mathrm{Hom}(N, N) = \Sigma_N$.*

Proof. Along a composite, the middle map y_* cancels,

$$\sigma(g \circ f) = z_*^{-1} \circ g'_* \circ f'_* \circ x_* = (z_*^{-1} \circ g'_* \circ y_*)(y_*^{-1} \circ f'_* \circ x_*) = \sigma(g)\,\sigma(f),$$

and

$$\sigma(\mathrm{id}) = x_*^{-1} \circ (\mathrm{id})_* \circ x_* = \mathrm{id}\,.$$

□

Lemma 5.10. *Let $f : (\bigvee a, x) \to (\bigvee b, y)$ in $\mathrm{Ch}^*(K)$ and suppose $\mathrm{blk}(x_*(k)) = \mathrm{blk}(x_*(l))$. Then $k < l$ if and only if $\sigma(f)(k) < \sigma(f)(l)$.*

Proof. Let $i = \mathrm{blk}(x_*(k))$, and write $f' : \bigvee a \to \bigvee b$ for the underlying wedge map of f. By Lemma 5.4, x is a tuple of runs, one per summand; let x_i be its i^{th} component. Likewise, let $j = \mathrm{blk}(f')(i)$, and y_j be the j^{th} component of y.

Now, since x is one dimensional, $\mathrm{blk} \circ x_* = \mathrm{blk}(x)$, which is monotone (Lemma 3.6). So $(\mathrm{blk} \circ x_*)^{-1}(i)$ is an interval, order-isomorphic to the positions $[a_i]$ of x_i, and likewise for y and $[b_j]$.

The morphism condition restricts to $x_i = \rho(f_i)(y_j)$ (Definition 3.4), so $(x_i)_* = (f_i)_*^{-1} \circ (y_j)_* \circ \iota$ by (4), where $\iota : [a_i] \to [b_j]$ is the increasing inclusion. Hence $\sigma(f) = y_*^{-1} \circ f'_* \circ x_*$ restricts to $(y_j)_*^{-1} \circ (f_i)_* \circ (x_i)_* = \iota$. Both identifications preserve order, so $k < l$ if and only if $\sigma(f)(k) < \sigma(f)(l)$. □

5.4 Comparison with the Salvetti poset

Theorem 5.11. *For every $n \geq 0$ there is an isomorphism of posets $\mathrm{Ch}^*(I^n) \cong \mathrm{Sal}(n)^{\mathrm{op}}$. Consequently $|\,\mathrm{Ch}^*(I^n)| \simeq \mathrm{Conf}(n, \mathbb{C})$ and $\pi_1|\,\mathrm{Ch}^*(I^n)| \cong P_n$.*

The proof is straightforward, although a bit tedious. Start with Lemma 5.1. Then send $\mathrm{Run}(I^n)$ to topes. For morphisms, the composition operator agrees with the projection. This is machine-checked in [Sto26].

6 The braiding

You'll notice that the K side is not involved in the computation of the permutation, or the proofs of its functoriality. We can formalize why this happens nicely.

Lemma 6.1. *Let Z be the terminal precubical set. Then $\mathrm{Ch}(Z) \cong \square^\vee$ and $\mathrm{Ch}^*(Z) \cong \square^\vee/\rho$, the comma category of serial wedges over ρ.*

Proof. Since Z is terminal, each serial wedge has exactly one bipointed map to it and every map of wedges commutes over it; so $W(Z) : \mathrm{Ch}(Z) \to \square^\vee$ is an isomorphism. Hence $\mathrm{Lines}(Z) = E(\rho)\circ W(Z)^{\mathrm{op}} = E(\rho)|_{\square^\vee}$. Taking categories of elements, $\mathrm{Ch}^*(Z) = \int \mathrm{Lines}(Z) = \int E(\rho)|_{\square^\vee} = \square^\vee/\rho$, the last step because the category of elements of $\mathrm{Hom}_\square(-,\rho)$ is the comma category. ☐

Lemma 6.2. Ch^* *is a functor* $\square_* \to \mathbf{Cat}$, *natural in* K *in the sense that* $\pi : \mathrm{Ch}^* \Rightarrow \mathrm{Ch}$ *is a natural transformation. Moreover, for every* K *the naturality square at the unique map* $!_K : K \to Z$ *to the final object,*

$$\begin{tikzcd}
\mathrm{Ch}^*(K) \arrow[r, "\mathrm{Ch}^*(!_K)"] \arrow[d, "\pi_K"'] \arrow[dr, phantom, "\lrcorner", very near start] & \mathrm{Ch}^*(Z) \arrow[d, "\pi_Z"] \\
\mathrm{Ch}(K) \arrow[r, "\mathrm{Ch}(!_K)"'] & \mathrm{Ch}(Z)
\end{tikzcd}$$

is a strict pullback in $\mathbf{Cat}$.

Proof. For $u : K \to L$, naturality of W gives $\mathrm{Lines}(L) \circ \mathrm{Ch}(u)^{\mathrm{op}} = E(\rho) \circ \big(W(L) \circ \mathrm{Ch}(u)\big)^{\mathrm{op}} = \mathrm{Lines}(K)$. So $\mathrm{Ch}^*(u)(c, x) := (\mathrm{Ch}(u)(c),\ x)$, and acts on morphisms via $\mathrm{Ch}(u)$. So we get functoriality and $\pi_L \circ \mathrm{Ch}^*(u) = \mathrm{Ch}(u) \circ \pi_K$.

Now take $u =!_K$, so $\mathrm{Lines}(K)$ is the restriction of $\mathrm{Lines}(Z)$ along $\mathrm{Ch}(!_K)$. Restricting a presheaf pulls back its category of elements, strictly: both $\mathrm{Ch}^*(K)$ and $\mathrm{Ch}(K) \times_{\mathrm{Ch}(Z)} \mathrm{Ch}^*(Z)$ have as objects the pairs $(c \in \mathrm{Ch}(K),\ x \in \mathrm{Lines}(Z)(\mathrm{Ch}(!_K)c))$, with the same morphisms. ☐

Note that K does not appear in the right-hand column: $\mathrm{Ch}^*(K)$ is determined by $\mathrm{Ch}(K)$ together with the single universal case $\mathrm{Ch}^*(Z)$.

6.1 Braid words from permutations

Now, a morphism f of $\mathrm{Ch}^*(K)$ has a permutation $\sigma(f)$, and it does not depend on K in the sense above. So it suffices to consider morphisms f of $\mathrm{Ch}^*(Z)$. But we still need a device for turning permutations into braid words. Which brings us to the Garside germ presentation of braids [Deh+15]. For any $n \in \mathbb{N}$, and $\sigma \in \Sigma_n$ write

$$\mathrm{Inv}(\sigma) := \{\, (i,j) \in [n] \times [n] \mid i < j \text{ and } \sigma(j) < \sigma(i) \,\}$$

for its set of *inversions*, and $\ell(\sigma) := |\,\mathrm{Inv}(\sigma)|$ for its *length*.

Definition 6.3. Let $\mathbf{Braid}^+$ be the category with objects the naturals and $\mathrm{Hom}(N,N) = B_N^+$, the positive braid monoid on N strands: generators $\sigma_1, \ldots, \sigma_{N-1}$ subject to the braid relations.

We want a different presentation: regard Σ_N as the Coxeter group of type A_{N-1}, so its Coxeter length is the ℓ above. Sending a reduced word for $\sigma \in \Sigma_N$ to its image in B_N^+ gives a well defined, injective map $\sigma \mapsto [\sigma]$, the *positive lift* [Deh+15, Lem. IX.1.22]. If $\ell(\sigma\sigma') = \ell(\sigma) + \ell(\sigma')$, then concatenating reduced words for σ and σ' gives one for $\sigma\sigma'$, so

$$[\sigma][\sigma'] = [\sigma\sigma'] \qquad \text{whenever} \qquad \ell(\sigma\sigma') = \ell(\sigma) + \ell(\sigma').$$

These relations in fact present [4] B_N^+, its *germ* presentation [Deh+15, Prop. IX.1.35].

[4] In the Lean, this fact takes Matsumoto's theorem as an axiom. Instead it defines braids by their germ presentation.

The key fact is a "no double crossings" feature of Ch^*: along a composite, no two events cross each other twice. Let's first state this in pure combinatorics.

Lemma 6.4. *Let f and g be permutations of $[N]$, and suppose that $\mathrm{Inv}(f) \subseteq \mathrm{Inv}(g \circ f)$. Then $\ell(g \circ f) = \ell(g) + \ell(f)$.*

Proof. Every element of $\mathrm{Inv}(g \circ f)$ either lies in $\mathrm{Inv}(f)$ or not, so

$$\mathrm{Inv}(g \circ f) \;=\; \big(\mathrm{Inv}(g \circ f) \cap \mathrm{Inv}(f)\big) \;\sqcup\; \big(\mathrm{Inv}(g \circ f) \setminus \mathrm{Inv}(f)\big).$$

By hypothesis $\mathrm{Inv}(f) \subseteq \mathrm{Inv}(g \circ f)$, so the first set equals $\mathrm{Inv}(f)$ and contributes $\ell(f)$.

For the second, a pair $k < l$ and $(k,l) \notin \mathrm{Inv}(f)$ has $f(k) < f(l)$, and then $(k,l) \in \mathrm{Inv}(g \circ f)$ iff $g(f(k)) > g(f(l))$, i.e. iff $(f(k), f(l)) \in \mathrm{Inv}(g)$. Since f is a bijection, $(p,q) \mapsto (f(p), f(q))$ restricts to a bijection

$$\mathrm{Inv}(g \circ f) \setminus \mathrm{Inv}(f) \;\xrightarrow{\sim}\; \{(p,q) \in \mathrm{Inv}(g) \;:\; f^{-1}(p) < f^{-1}(q)\}.$$

This target is all of $\mathrm{Inv}(g)$: otherwise take $(p,q) \in \mathrm{Inv}(g)$ and set $k = f^{-1}(q) < l = f^{-1}(p)$. Then $f(k) = q > p = f(l)$ gives $(k,l) \in \mathrm{Inv}(f)$, while $g(f(k)) = g(q) < g(p) = g(f(l))$ gives $(k,l) \notin \mathrm{Inv}(g \circ f)$, contradicting $\mathrm{Inv}(f) \subseteq \mathrm{Inv}(g \circ f)$. Hence the second set contributes $\ell(g)$, and $\ell(g \circ f) = \ell(f) + \ell(g)$. ☐

Lemma 6.5 (crossings are intra-block)**.** *Let $g : (\bigvee c \to K, r) \to (\bigvee b \to K, q)$ in $\mathrm{Ch}^*(K)$ and $(k,l) \in \mathrm{Inv}(\sigma(g))$. Then $(\mathrm{blk} \circ q_* \circ \sigma(g))(k) = (\mathrm{blk} \circ q_* \circ \sigma(g))(l)$.*

Proof. Put $u := \sigma(g)(k)$ and $v := \sigma(g)(l)$, and $g' : \bigvee c \to \bigvee b$ be the induced wedge map. So $k < l$ and $v < u$. Since $q_* \circ \sigma(g) = g'_* \circ r_*$ we have $q_*(u) = g'_*(r_*(k))$ and $q_*(v) = g'_*(r_*(l))$, so by Definition 3.4 and the monotonicity of $\mathrm{blk}(g')$ and of $\mathrm{blk} \circ r_* = \mathrm{blk}(r)$ (Lemma 3.6),

$$\mathrm{blk}(q_*(u)) = \mathrm{blk}(g')\big(\mathrm{blk}(r_*(k))\big) \;\leq\; \mathrm{blk}(g')\big(\mathrm{blk}(r_*(l))\big) = \mathrm{blk}(q_*(v)),$$

using $k < l$. On the other hand $v < u$ and monotonicity of $\mathrm{blk} \circ q_* = \mathrm{blk}(q)$ (Lemma 3.6) give $\mathrm{blk}(q_*(v)) \leq \mathrm{blk}(q_*(u))$. Hence $\mathrm{blk}(q_*(u)) = \mathrm{blk}(q_*(v))$. ☐

Theorem 6.6. *The map $f \mapsto [\sigma(f)]$ yields a functor $\mathrm{Conc} : \mathrm{Ch}^*(K) \to \mathbf{Braid}^+$, natural in K.*

Proof. An object goes to the braid object on its number of coordinates, well defined by Lemma 3.6, and $\sigma(\mathrm{id}) = \mathrm{id}$. For composition we need $[\sigma(f)][\sigma(g)] = [\sigma(f) \circ \sigma(g)]$, which by the germ presentation holds once $\ell(\sigma(f) \circ \sigma(g)) = \ell(\sigma(f)) + \ell(\sigma(g))$, and by Lemma 6.4 it is enough to prove $\mathrm{Inv}(\sigma(g)) \subseteq \mathrm{Inv}(\sigma(f) \circ \sigma(g))$ for composable $g : (\bigvee c, r) \to (\bigvee b, q)$ and $f : (\bigvee b, q) \to (\bigvee a, p)$.

Let $(k,l) \in \mathrm{Inv}(\sigma(g))$ and put $u := \sigma(g)(k)$, $v := \sigma(g)(l)$, so $v < u$. By Lemma 6.5, $(\mathrm{blk} \circ q_*)(u) = (\mathrm{blk} \circ q_*)(v)$, so Lemma 5.10 applied to f turns $v < u$ into $\sigma(f)(v) < \sigma(f)(u)$, which is exactly $(k,l) \in \mathrm{Inv}(\sigma(f \circ g)) = \mathrm{Inv}(\sigma(f) \circ \sigma(g))$, the last equality by Lemma 5.9.

For naturality in K, note that σ factors through $\mathrm{Ch}^*(Z)$, so $\mathrm{Conc}(K) = \mathrm{Conc}(Z) \circ \mathrm{Ch}^*(!_K)$ by Lemma 6.2. For a general $u : K \to L$ we have $!_L \circ u = !_K$, hence $\mathrm{Ch}^*(!_L) \circ \mathrm{Ch}^*(u) = \mathrm{Ch}^*(!_K)$ and so $\mathrm{Conc}(L) \circ \mathrm{Ch}^*(u) = \mathrm{Conc}(K)$. ☐

Proposition 6.7. *For any bipointed precubical set K, if K contains a 2-cell, then $\mathrm{Conc}(K)$ is nontrivial.*

Proof. First, note that if K has a 2-cell, then there is a map $u : I^2 \to K$. Then, by naturality

$$\begin{tikzcd} \mathrm{Ch}^*(I^2) \arrow[d, "\mathrm{Ch}^*(u)"] \arrow[dr, "\mathrm{Conc}(I^2)"] & \\ \mathrm{Ch}^*(K) \arrow[r, "\mathrm{Conc}(K)"'] & \mathbf{Braid}^+ \end{tikzcd}$$

Consider the map $p : \bigvee[1,1] \to \bigvee[2]$ going along the bottom edge and then the right edge, and the map $q : \bigvee[1,1] \to \bigvee[2]$ going along the left edge and then the top edge. We have

$$\begin{tikzcd} \bigvee[1,1] \arrow[r, "\mathrm{id}"] \arrow[d, dashed, "\sigma(p)"'] & \bigvee[1,1] \arrow[d, "p"] \arrow[dr] & \\ \bigvee[1,1] \arrow[r, "q"'] & \bigvee[2] \arrow[r, "f"'] & I^2 \end{tikzcd}$$

This is a morphism of $\mathrm{Ch}^*(I^2)$ from $(\bigvee[1,1] \xrightarrow{f \circ p} I^2,\ \mathrm{id})$ to $(\bigvee[2] \xrightarrow{f} I^2,\ q)$: the condition $x = \mathrm{Lines}(p)(y)$ is automatic because $\mathrm{Run}(\bigvee[1,1])$ is a singleton. Its permutation is

$$\sigma(p) = q_*^{-1} \circ p_* \circ \mathrm{id} = \{1 \mapsto 2,\ 2 \mapsto 1\},$$

because p traverses direction 1 then direction 2 while q traverses direction 2 then direction 1. This has an inversion, so $[\sigma(p)]$ is not the identity braid. □

7 Next steps

1. **What does $|\mathrm{Ch}^*(K)|$ compute?** It appears that Conc is a measurement of the concurrency structure of K, but it's unclear exactly what it computes in general. For Euclidean complexes, it seems to have slightly less vague semantics, since every strand corresponds to an axis. While $\mathrm{Ch}^*(K)$ carries as much information as $\mathrm{Ch}(K)$, Conc seems to depend only on the 3-skeleton.
2. **A chain complex.** The category $\mathrm{Ch}^*(K)$ is graded by the shape of the wedges: (the sum of the dimensions) − (the length of the list). Each morphism should decompose into a sequence of degree 1 morphisms. More work is required to see what kinds of properties this has.
3. **Sensitivity.** Using the verified computations in Lean, we have established several features. Firstly, the image of Conc seems insensitive to the topology. Both I^3 and the boundary of I^3 (that is, I^3 without its top 3-cell) produce the whole of P_n. However the kernels of Conc are quite distinct, so Conc itself is sensitive to the topology. The degree 1 wedges determine which generators occur, and the degree 2 wedges determine which Artin relations are faithful.
4. **Braid Closures** When looking at loops in the free groupoid of $\mathrm{Ch}^*(K)$, it makes intuitive sense to consider not just the braid, but the braid closure. This gives us a device for adding new strands to the braids, via the Markov move. There is also a desire to build something invariant under subdivision, which also adds new strands. Perhaps these goals align? Furthermore, it may give a nice braid theoretic explanation of why only the 3-skeleton occurs, as knots are all trivial in codimension 3 and above.

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